MYSQL PROJECT

TOPIC: INSURANCE COST ANALYSIS USING MYSQL

SOURCE: KAGGLE

INTRODUCTION

This project performs a structured analysis of a medical insurance dataset using MySQL. The dataset contains customer-level information for 1,338 customers, including age, sex, BMI, number of children, smoking status, region, and the individual insurance charge billed.

The objective of this analysis is to find out which factors are associated with insurance charges, such as smoking, age, BMI, family size, region and gender, using SQL techniques such as aggregation, GROUP BY, CASE statements, subqueries, CTEs, and window functions.

All charges in the dataset are in US dollars.

TOOL: MySQL Workbench

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Queries

Database setup

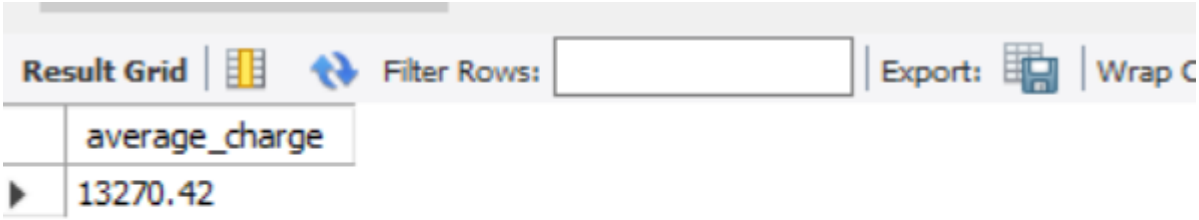
```
CREATE DATABASE insurance;

USE insurance;

SELECT * FROM insurance;
```

1. Calculate the overall average insurance charge

```
SELECT ROUND(AVG(charges), 2) AS average_charge
FROM insurance;
```



average_charge
13270.42

Insight: The average insurance charge across all customers is about 13,270. This is the baseline number used to compare every group in the queries that follow.

2. Compare the average charge of smokers vs non-smokers

```
SELECT
  smoker,
  COUNT(*) AS customers,
  ROUND(AVG(charges), 2) AS average_charge
FROM insurance
GROUP BY smoker;
```

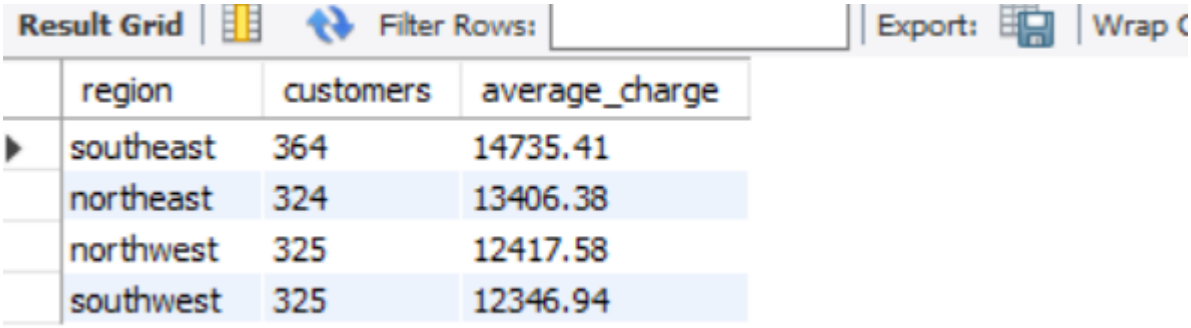


smoker	customers	average_charge
yes	274	32050.23
no	1064	8434.27

Insight: Smokers (274 customers) pay an average of 32,050, while non-smokers (1,064 customers) pay 8,434. Smokers pay roughly 3.8 times more, making smoking status the strongest cost factor so far.

3. Find the average charge by region

```
SELECT
  region,
  COUNT(*) AS customers,
  ROUND(AVG(charges), 2) AS average_charge
FROM insurance
GROUP BY region
ORDER BY average_charge DESC;
```



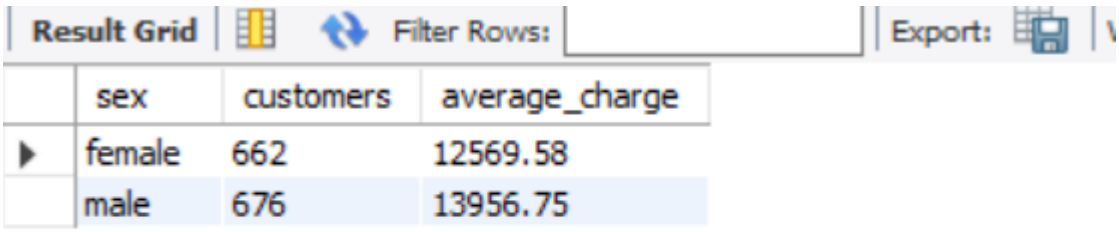
Result Grid | Filter Rows: | Export: | Wrap

region	customers	average_charge
southeast	364	14735.41
northeast	324	13406.38
northwest	325	12417.58
southwest	325	12346.94

Insight: The southeast has the highest average charge (14,735) and also the most customers (364). The southwest is lowest at 12,347. Regional differences exist but are fairly small compared to the smoker gap.

4. Find the average charge by gender

```
SELECT
  sex,
  COUNT(*) AS customers,
  ROUND(AVG(charges), 2) AS average_charge
FROM insurance
GROUP BY sex;
```



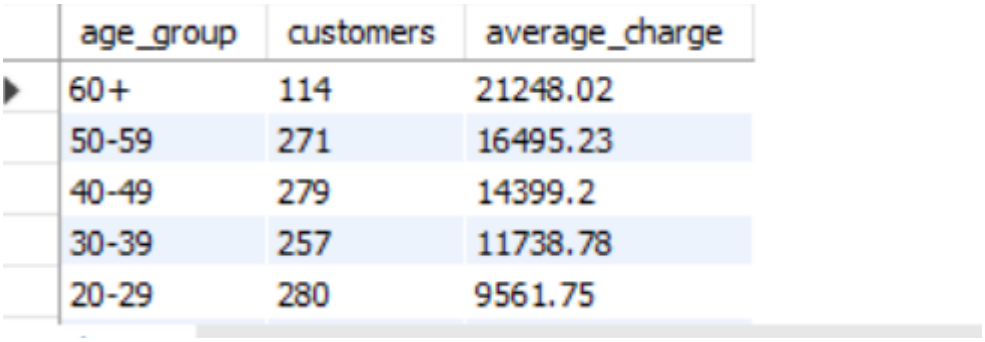
Result Grid | Filter Rows: | Export:

sex	customers	average_charge
female	662	12569.58
male	676	13956.75

Insight: Males pay about 13,957 on average and females about 12,570, a gap of roughly 11%. The customer split is almost even (676 male, 662 female), but Query 8 shows the gap comes mainly from smoking: more males are smokers (159 vs 115), and smokers pay far more.

5. Find the average charge by age group

```
SELECT
  CASE
    WHEN age < 20 THEN 'Under 20'
    WHEN age BETWEEN 20 AND 29 THEN '20-29'
    WHEN age BETWEEN 30 AND 39 THEN '30-39'
    WHEN age BETWEEN 40 AND 49 THEN '40-49'
    WHEN age BETWEEN 50 AND 59 THEN '50-59'
    ELSE '60+'
  END AS age_group,
  COUNT(*) AS customers,
  ROUND(AVG(charges), 2) AS average_charge
FROM insurance
GROUP BY age_group
ORDER BY average_charge DESC;
```



age_group	customers	average_charge
60+	114	21248.02
50-59	271	16495.23
40-49	279	14399.2
30-39	257	11738.78
20-29	280	9561.75

Insight: Charges rise steadily with age. Customers aged 60+ pay about 21,248 on average, over twice the 9,562 paid by those aged 20-29. Age is a clear and consistent cost driver.

6. Find the average charge by BMI category

```
SELECT
  CASE
    WHEN bmi < 18.5 THEN 'Underweight'
    WHEN bmi < 25 THEN 'Normal'
    WHEN bmi < 30 THEN 'Overweight'
    ELSE 'Obese'
  END AS bmi_category,
  COUNT(*) AS customers,
  ROUND(AVG(charges), 2) AS average_charge
FROM insurance
GROUP BY bmi_category
ORDER BY average_charge DESC;
```

Result Grid | Filter Rows: | Expo

bmi_category	customers	average_charge
Obese	707	15552.34
Overweight	386	10987.51
Normal	225	10409.34
Underweight	20	8852.2

Insight: Obese customers have the highest average charge (15,552) and make up more than half of all customers (707 of 1,338). Overweight, Normal and Underweight groups all sit between about 8,850 and 10,990.

7. Find the average charge by number of children

```
SELECT
  children,
  COUNT(*) AS customers,
  ROUND(AVG(charges), 2) AS average_charge
FROM insurance
GROUP BY children
ORDER BY children;
```

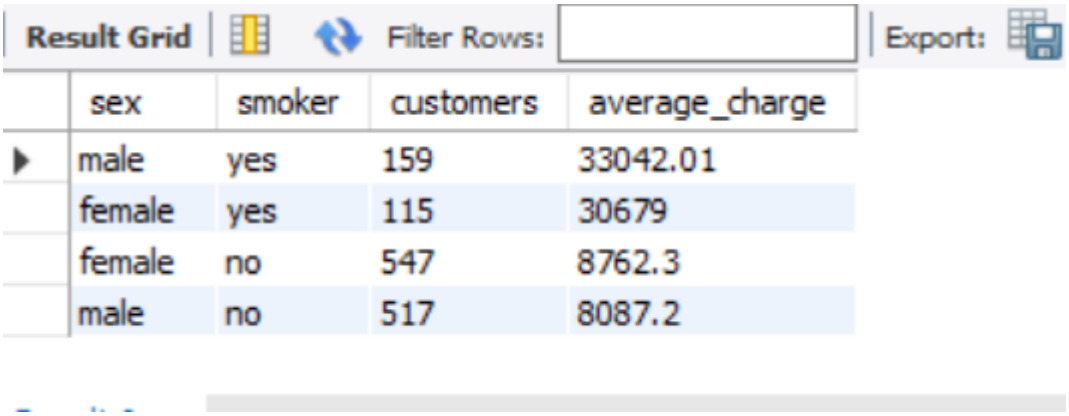
Result Grid | Filter Rows: | Exp

children	customers	average_charge
0	574	12365.98
1	324	12731.17
2	240	15073.56
3	157	15355.32
4	25	13850.66

Insight: Average charges generally increase as the number of children goes up, from about 12,366 for no children to 15,355 for three children. The dip at four children is based on only 25 customers, so it is less reliable.

8. Find the average charge by gender and smoking status

```
SELECT
  sex,
  smoker,
  COUNT(*) AS customers,
  ROUND(AVG(charges), 2) AS average_charge
FROM insurance
GROUP BY sex, smoker
ORDER BY average_charge DESC;
```



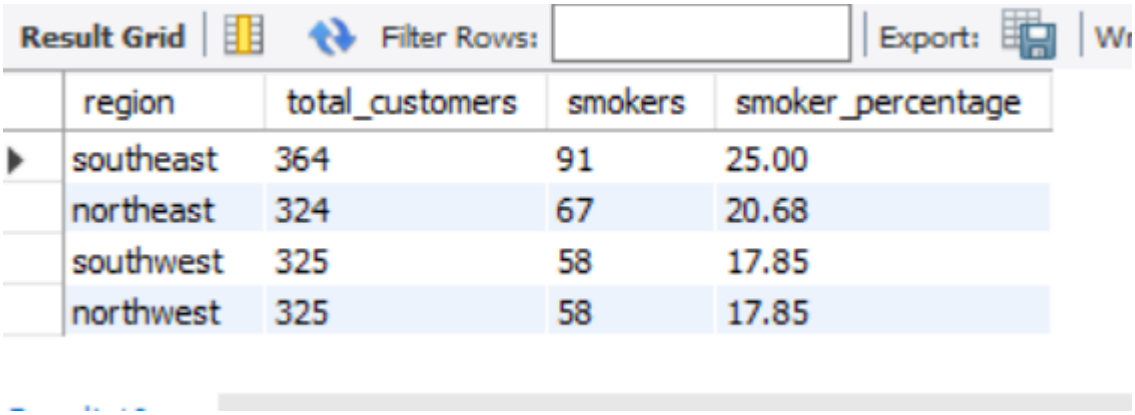
Result Grid | Filter Rows: | Export:

sex	smoker	customers	average_charge
male	yes	159	33042.01
female	yes	115	30679
female	no	547	8762.3
male	no	517	8087.2

Insight: Smoking status matters far more than gender. Male smokers pay 33,042 and female smokers 30,679, while non-smokers pay only about 8,000-8,800 for both genders. Males cost more overall (Query 4) mainly because more of them smoke (159 vs 115).

9. Calculate the smoking percentage by region

```
SELECT
  region,
  COUNT(*) AS total_customers,
  SUM(CASE WHEN smoker = 'yes' THEN 1 ELSE 0 END) AS smokers,
  ROUND(
    SUM(CASE WHEN smoker = 'yes' THEN 1 ELSE 0 END) * 100.0
    / COUNT(*), 2
  ) AS smoker_percentage
FROM insurance
GROUP BY region
ORDER BY smoker_percentage DESC;
```



Result Grid | Filter Rows: | Export:

region	total_customers	smokers	smoker_percentage
southeast	364	91	25.00
northeast	324	67	20.68
southwest	325	58	17.85
northwest	325	58	17.85

Insight: The southeast has the highest share of smokers at 25.00%, while the northwest and southwest are both at 17.85%. This links to Query 3: the region with the most smokers also has the highest average charge.

10. List the top 10 highest insurance charges

```
SELECT
  age,
  sex,
  bmi,
  children,
  smoker,
  region,
  ROUND(charges, 2) AS charges
FROM insurance
ORDER BY charges DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap

age	sex	bmi	children	smoker	region	charges
54	female	47.41	0	yes	southeast	63770.43
45	male	30.36	0	yes	southeast	62592.87
52	male	34.485	3	yes	northwest	60021.4
31	female	38.095	1	yes	northeast	58571.07
33	female	35.53	0	yes	northwest	55135.4

Result 11

Insight: Every customer in the visible top rows is a smoker, and each has a BMI above 30. The highest charge is 63,770, nearly 5 times the overall average. Extreme charges are strongly tied to smoking combined with a high BMI.

11. Find the percentage of smokers and non-smokers

```
SELECT
  smoker,
  COUNT(*) AS customers,
  ROUND(
    COUNT(*) * 100.0 / (SELECT COUNT(*) FROM insurance),
    2
  ) AS percentage
FROM insurance
GROUP BY smoker;
```

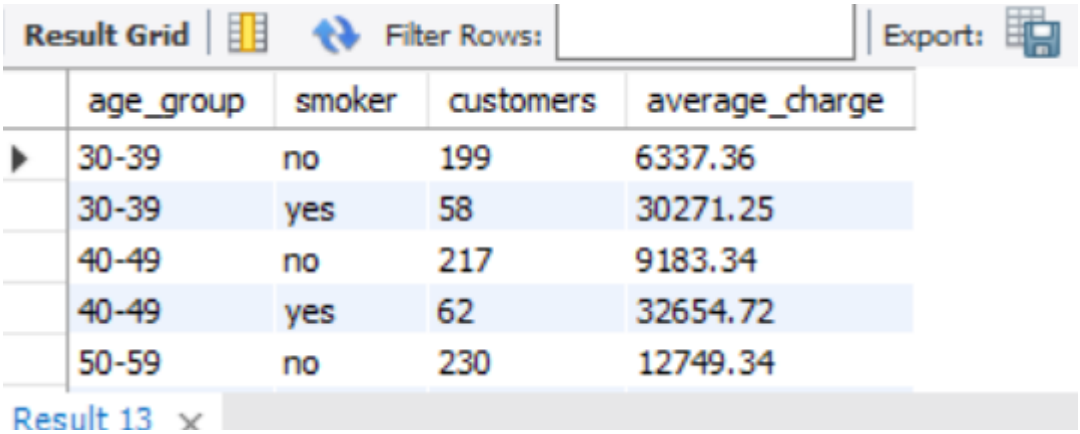
Result Grid | Filter Rows: | Ex

smoker	customers	percentage
yes	274	20.48
no	1064	79.52

Insight: Only 20.48% of customers are smokers, while 79.52% are non-smokers. A small group of customers is responsible for a large share of the high charges.

12. Analyse the average charge by age group and smoking status

```
SELECT
  CASE
    WHEN age < 30 THEN 'Under 30'
    WHEN age BETWEEN 30 AND 39 THEN '30-39'
    WHEN age BETWEEN 40 AND 49 THEN '40-49'
    WHEN age BETWEEN 50 AND 59 THEN '50-59'
    ELSE '60+'
  END AS age_group,
  smoker,
  COUNT(*) AS customers,
  ROUND(AVG(charges), 2) AS average_charge
FROM insurance
GROUP BY age_group, smoker
ORDER BY age_group, smoker;
```



age_group	smoker	customers	average_charge
30-39	no	199	6337.36
30-39	yes	58	30271.25
40-49	no	217	9183.34
40-49	yes	62	32654.72
50-59	no	230	12749.34

Result 13

Insight: In every age group, smokers pay far more than non-smokers. At ages 30-39, smokers pay 30,271 versus 6,337 for non-smokers (about 4.8 times). At ages 40-49, the gap is 32,655 versus 9,183 (about 3.6 times). Smoking has a bigger effect than age.

13. Analyse the average charge by BMI category and smoking status

```
SELECT
  CASE
    WHEN bmi < 18.5 THEN 'Underweight'
    WHEN bmi < 25 THEN 'Normal'
    WHEN bmi < 30 THEN 'Overweight'
    ELSE 'Obese'
  END AS bmi_category,
  smoker,
  COUNT(*) AS customers,
  ROUND(AVG(charges), 2) AS average_charge
FROM insurance
GROUP BY bmi_category, smoker
ORDER BY average_charge DESC;
```

Result Grid | Filter Rows: | Export:

bmi_category	smoker	customers	average_charge
Obese	yes	145	41557.99
Overweight	yes	74	22495.87
Normal	yes	50	19942.22
Underweight	yes	5	18809.82
Obese	no	562	8842.69

Insight: Obese smokers have the highest average charge at 41,558, which is about 4.7 times the 8,843 paid by obese non-smokers. BMI alone raises costs only modestly, but high BMI combined with smoking raises them sharply.

14. Count customers with charges above the overall average

```
SELECT
  COUNT(*) AS customers_above_average
FROM insurance
WHERE charges > (
  SELECT AVG(charges)
  FROM insurance
);
```

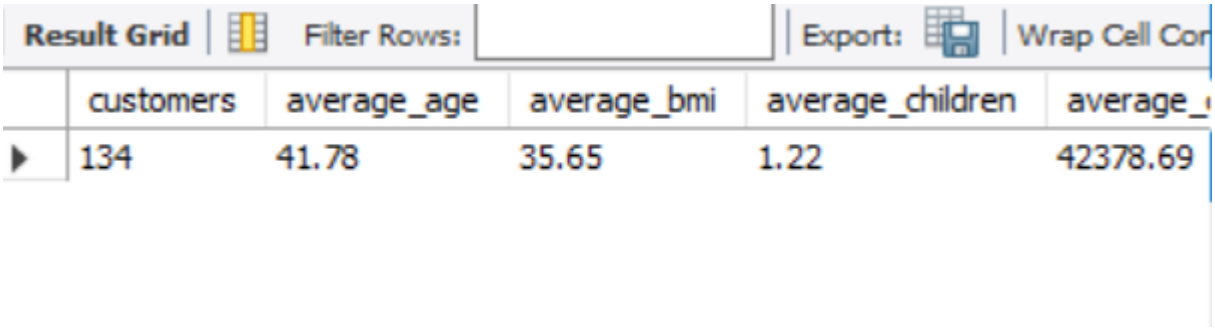
Result Grid | Filter Rows: | Export:

customers_above_average
420

Insight: Only 420 of 1,338 customers (about 31%) are above the average charge. This means the average is pulled upward by a minority of very high charges, so most customers pay less than the average.

15. Profile the top 10% of customers by insurance charge

```
WITH ranked_customers AS (
  SELECT
    *,
    NTILE(10) OVER (ORDER BY charges DESC) AS charge_group
  FROM insurance
)
SELECT
  COUNT(*) AS customers,
  ROUND(AVG(age), 2) AS average_age,
  ROUND(AVG(bmi), 2) AS average_bmi,
  ROUND(AVG(children), 2) AS average_children,
  ROUND(AVG(charges), 2) AS average_charge,
  SUM(CASE WHEN smoker = 'yes' THEN 1 ELSE 0 END) AS smokers
FROM ranked_customers
WHERE charge_group = 1;
```



customers	average_age	average_bmi	average_children	average_
134	41.78	35.65	1.22	42378.69

Insight: The top 10% group has 134 customers with an average charge of 42,379, over 3 times the overall average. Their average BMI of 35.65 is in the obese range and their average age is about 42. This group is the main target for cost and risk management.

CONCLUSION

The Insurance Cost Analysis project used MySQL to study what is associated with medical insurance charges. Using aggregation, GROUP BY, CASE statements, subqueries, a CTE, and a window function (NTILE), the raw customer records were turned into clear insights.

Smoking showed the strongest association with high charges: smokers pay about 3.8 times more than non-smokers, and this gap holds across every age group, BMI category and gender. Age also raises charges steadily, and BMI has its biggest impact when combined with smoking, with obese smokers paying the most. Region, gender and number of children showed smaller differences.

Only about 31% of customers pay more than the average, and the top 10% pay over three times the average, so a small, high-risk group accounts for much of the total cost. These findings show associations in this dataset, not proven causes, since factors like age, BMI and smoking overlap.

Overall, this project strengthened practical SQL skills and showed how data analysis can support pricing, risk assessment and decision-making in the insurance domain.